112-1 Andriod App 期末報告

＜MEATING＞

GitHub 專用

指導老師：[redacted]

組別：第五組

組長：資管[redacted]司[redacted]略 kavisszeto

組員：資管三甲[redacted]梁[redacted]敏

資管三乙[redacted]李[redacted]賢

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中華民國　113 年　1 月　11 日

一.內容

a. App 設計需求

此應用程式主要是為傳統即食肉品企業設計。由於在現今科技當道的世代中，線上購物早已成趨勢，令一般傳統品牌的地位受到挑戰。特別是經過 2019 年末開始的全球性瘟疫，電商平台突破疫情障礙，其業績成長了 16.1%。

因此，我們希望開發一個應用程式，讓顧客可以從中購物之外，還能接觸到企業的相關資訊，如查閱推廣海報和傳送訊息至企業。配合會員系統及其儲值功能，可讓企業掌握顧客行為及偏好，提高顧客滿意度。「MEATING」為我們的假想企業，此名稱將會出現在以下各種示意圖中。

功能性需求

✧ 購物系統 ➢ 商店 ➢ 購物車 ✧ 電子錢包 ➢ 儲值 ➢ 付款 ➢ 查看消費記錄 ✧ 會員 ➢ 註冊 ➢ 登入 ➢ 修改個人資料 ➢ 登出 ✧ 查看最新消息 ✧ 查看分店位置 ✧ 查詢企業更多資訊 ✧ 電郵傳送意見 ✧ 推送提示	已完成功能 1. 啟動 icon (launch icon) 2. 基本會員登入操作功能 3. 主畫面 - Navigation Drawer、選單列 (menu bar)、浮動按鈕 (FAB)、Toast 訊息、對話框(Dialogs)、下拉選單 (Spinner)、捲動頁面 (Scrolling) 4. 訊息通知功能： Notification, Broadcast Receiver 5. 資料儲存功能： Shared Preference, SQLite Database

b. 功能示意圖



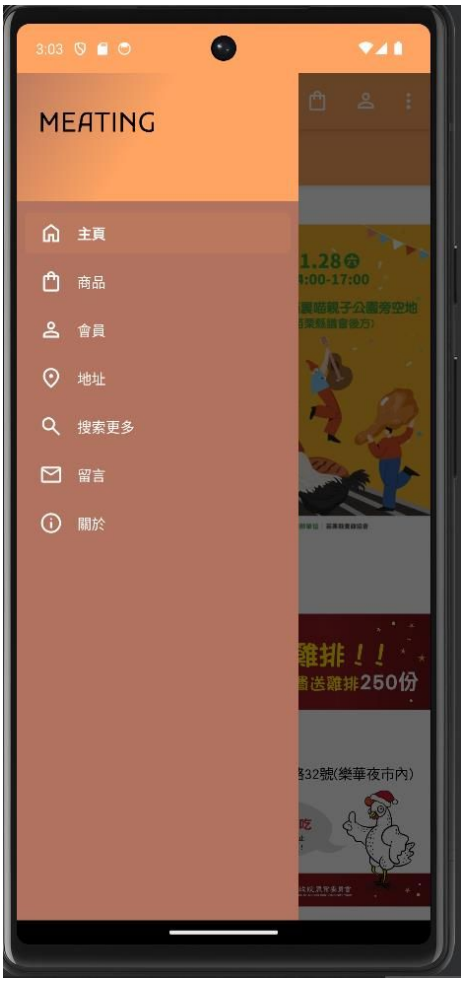
應用程式圖示

以紅框所圈着的應用程式「Meating」為本假想企業的圖示。本組更改圖示為肉品，呼應本應用程式是為即食肉品企業所設計。使用者按下圖示按鈕便可進入應用程式中使用各種功能。



應用程式主頁

進入到應用程式後，首先打開的就是我們的主頁。正中央為最新資訊，顯示企業近期推廣資訊或外部公告。使用者可以上下滑動瀏覽更多資訊。由於沒有為假想企業設計海報或公告，所以海報圖為「國產咕咕 Good」所設計的。左上方菜單圖示按鈕可打開菜單，瀏覽其他功能。購物袋圖示按鈕為瀏覽商品，顧客可進入商品頁面選購商品。購物袋圖示按鈕右邊的人像圖示按鈕為會員登入之處，按進後便可進入登入畫面。最後右上三點圖示按鈕為其他功能的菜單，分別可查詢地址、搜尋網絡、連接 Gmail 讓顧客留言和查看公司簡介（現為顯示本組的組員名單）。



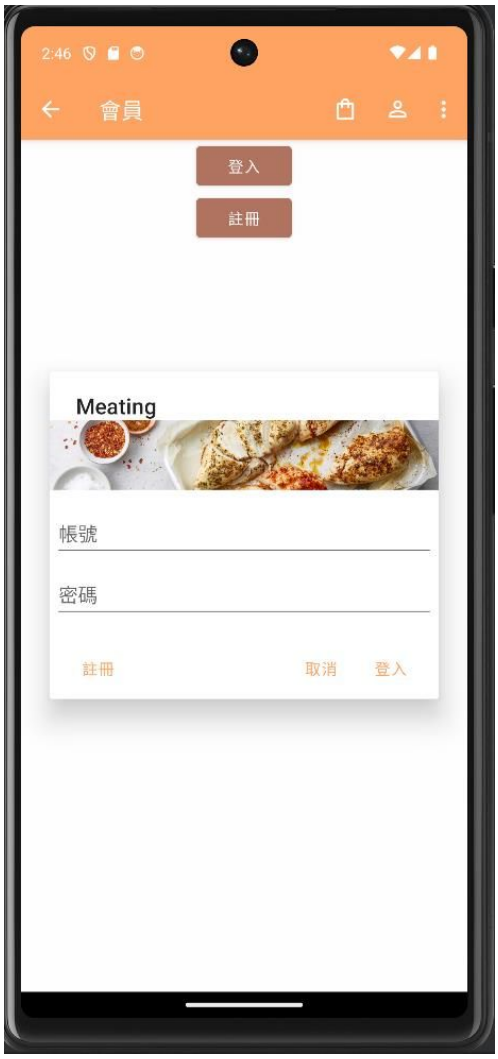
按下導覽圖示按鈕後顯示的菜單

左上角顯示了本假想企業的標誌，其後的深色部分為菜單目錄。以上菜單內容的圖示或文字按下後都會跳進去相關畫面，以上圖示按鈕都對應到相關功能。屋子圖示按鈕為進入主頁，商品圖示按鈕為瀏覽商品，人像圖示按鈕為會員登入之處，定位圖示按鈕為查詢地址，放大鏡圖示按鈕為搜尋網絡，信封圖示按鈕為連接 Gmail 讓顧客留言，以及「i」圖示按鈕為查看公司簡介。



按下會員圖示按鈕後的會員介面

登入按鈕為現有會員登入之處，非會員可透過按下註冊按鈕完成註冊流程並獲得會員帳戶。



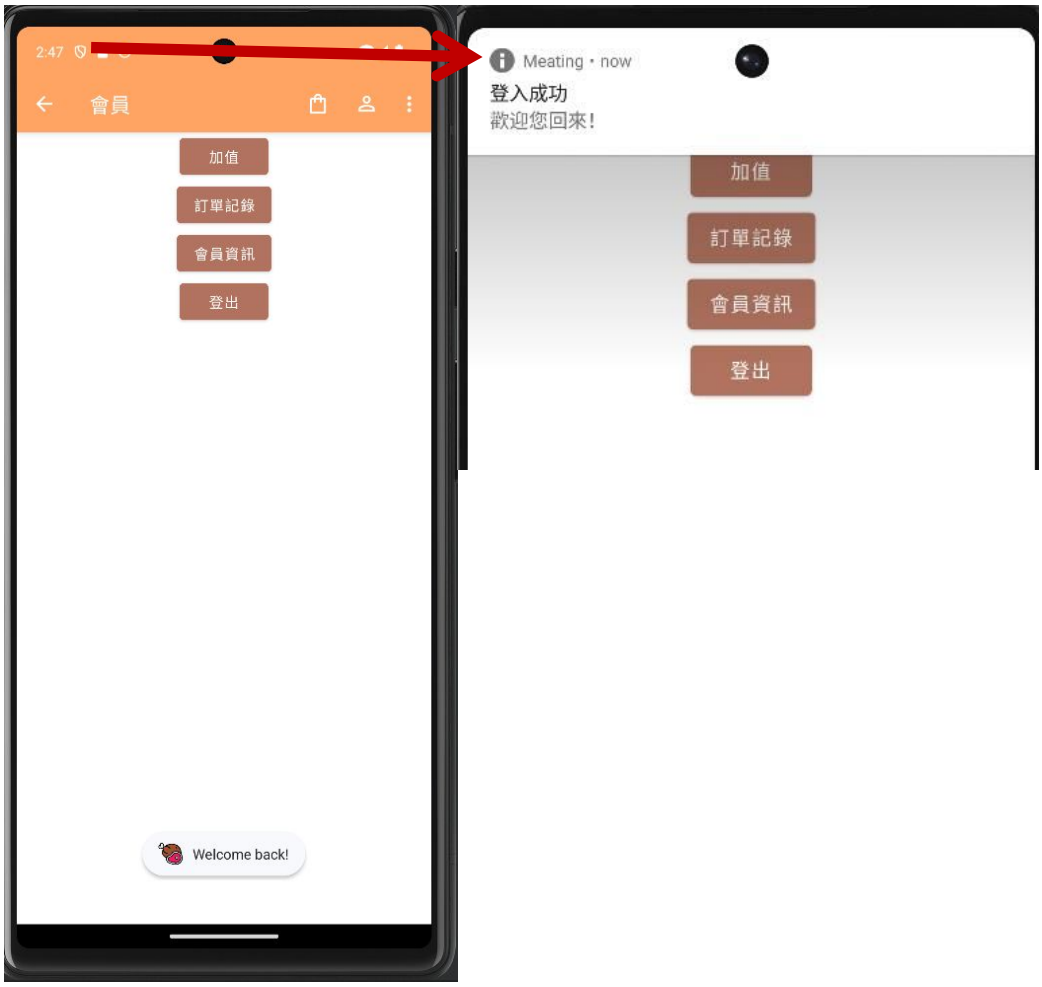
按下登入按鈕後跳出會員登入畫面

會員只需在相應的欄位中輸入帳號及其密碼後，便可按下登入按鈕登入會員系統進行購物。非會員發現自己沒有相關帳號後亦可按下左下角註冊按鈕進行註冊程序。取消按鈕即為終止目前登入程序，回到會員介面。



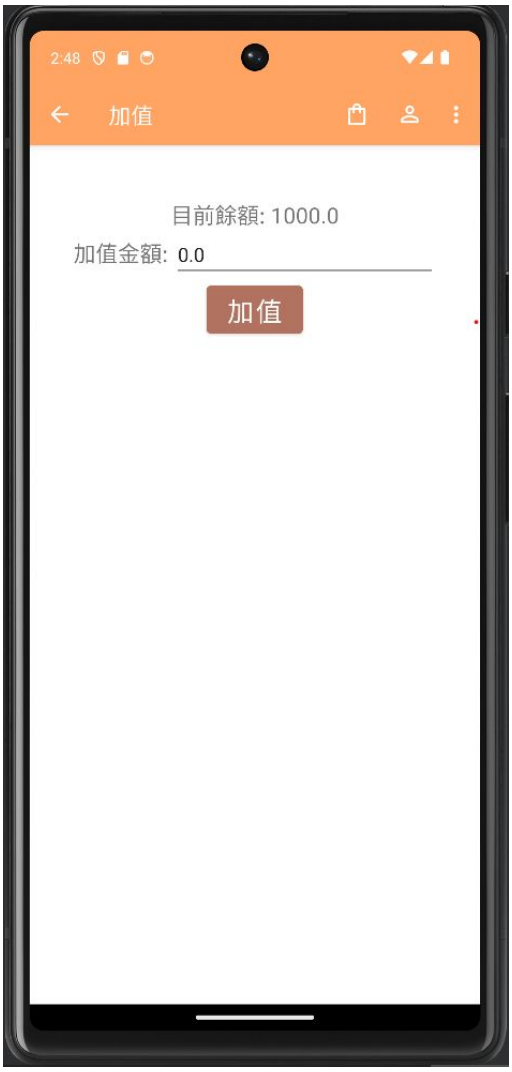
按下註冊按鈕後的會員註冊畫面

非會員使用者可透過填寫下面欄位註冊新的帳戶。除了創作一個用戶帳號名稱之外，還有創作一組密碼用於登入其專屬帳號。為了避免錯誤，使用者需重新輸入與上列一樣的密碼，來確認與所創作密碼是一致的。



登入會員後的會員介面

登入會員後，會跳出登入成功提示，會員可於會員介面中進行一系列的操作。按下加值按鈕後，會員可儲值金額於應用程式內使用，可方便消費者直接購買企業相關產品。按下訂單紀錄按鈕後，會員可查詢過往所下過的訂單資訊。按下會員資訊按鈕後，會員即可瀏覽自己所設置的個人資料及相關資訊。最後按下登出按鈕後，會員即登出目前所登入的帳戶，可讓使用者登入其他帳號。



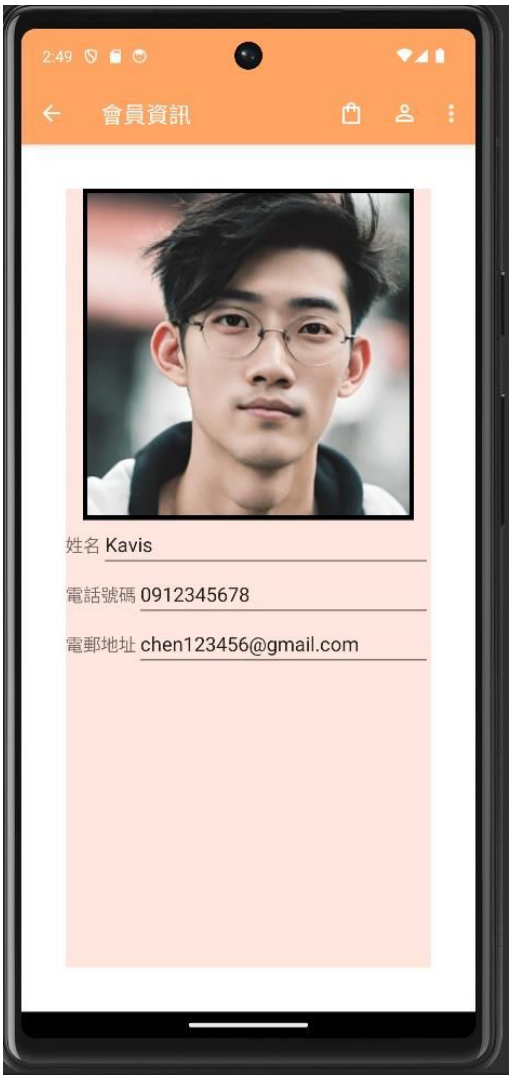
按下加值按鈕後的加值畫面

顧客只需填寫想加值的金額於欄位上，再按下加值按鈕後便可成功加值。目前系統沒有連接支付平台，因此輸入數值並按下按鍵後便會立即加值。



按下訂單紀錄按鈕後的畫面

會員可透過此頁面瀏覽最近的訂單紀錄，訂單紀錄以倒序的方式顯示最新訂單編號、訂單金額、收件人姓名、手機號碼、地址。



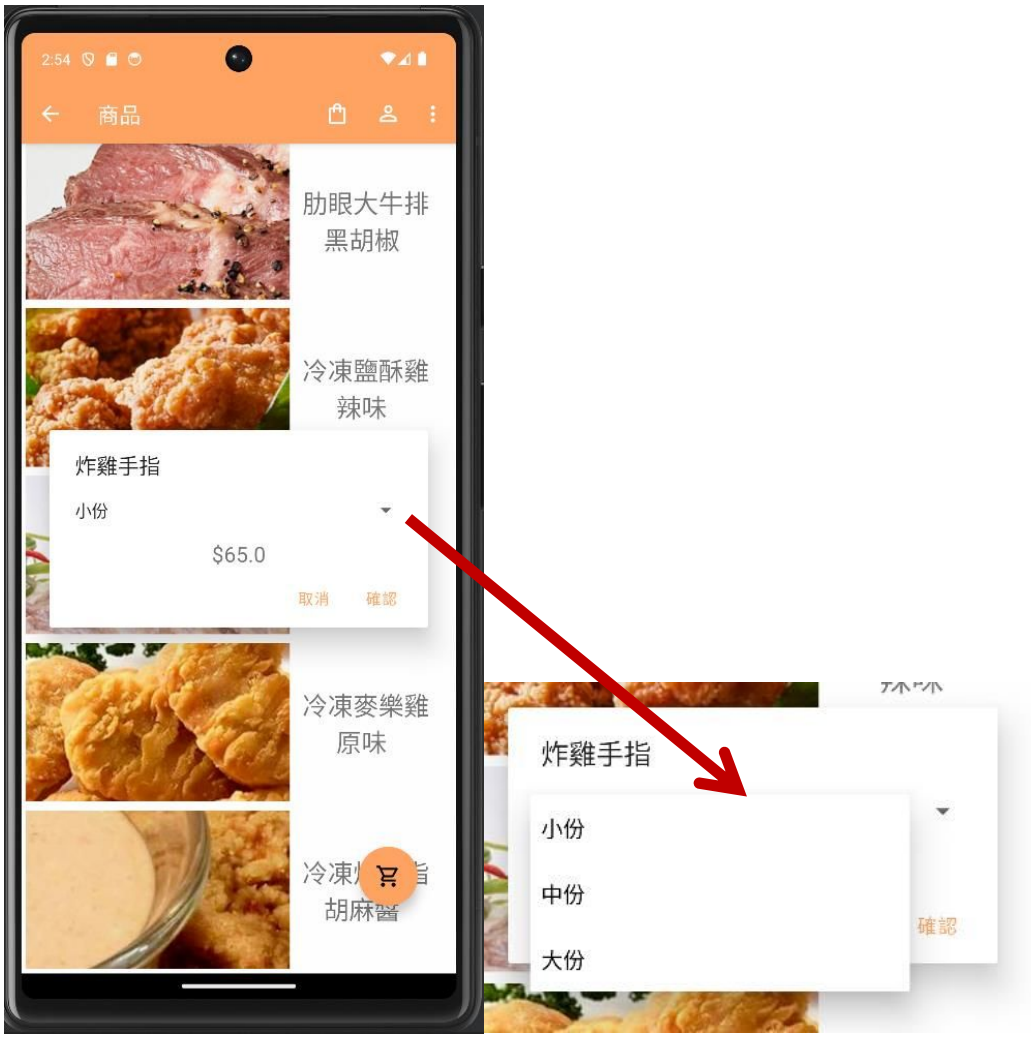
按下會員資訊按鈕後的畫面

會員可透過此頁面查詢會員自己所登記的資料，分別顯示了會員姓名、電機號碼，以及電郵地址。



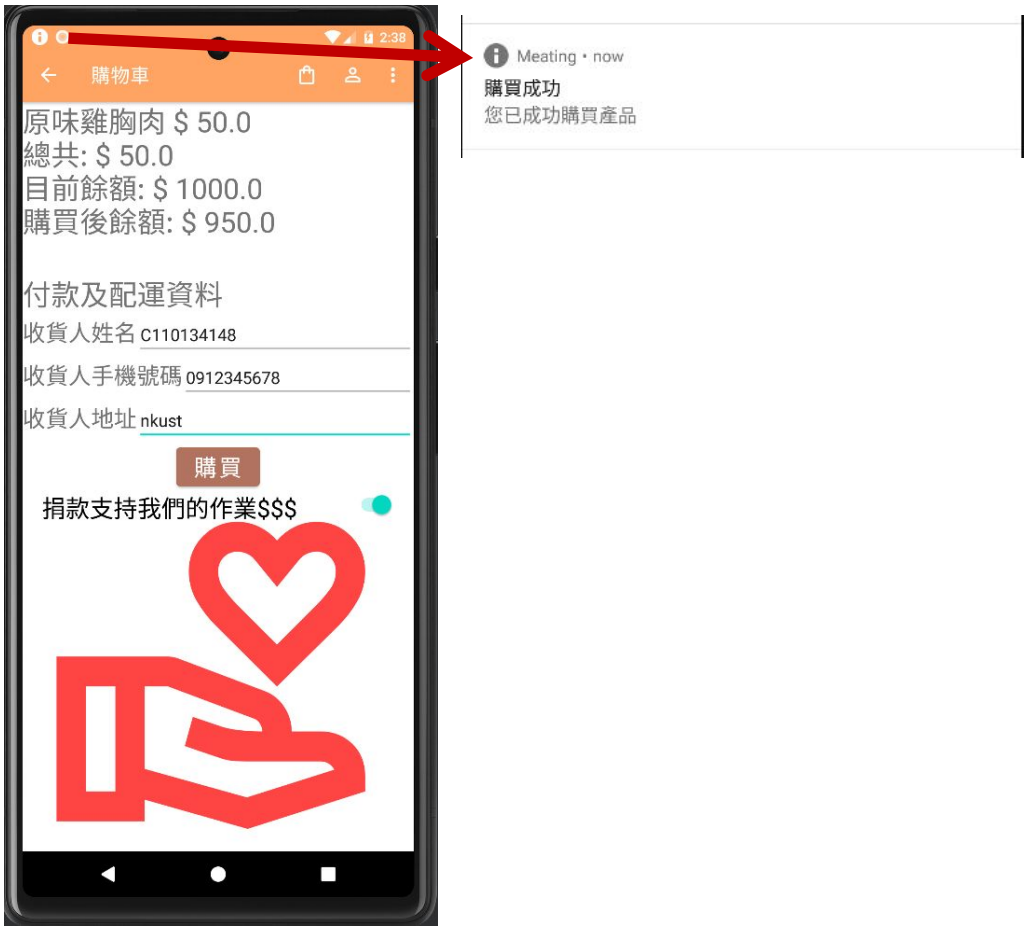
按下購物袋圖示按鈕後的商品畫面

使用者進入此頁面後，可以上下滑動瀏覽商品。若使用者想購買頁面中的商品，他們只需按下產品圖片或文字便可進行下一步挑選產品其他選項。購物車圖示按鈕為購物車功能，使用者可按下後即可瀏覽已放進購物車的商品。



按下商品圖片及菜單選項功能鍵後的畫面

按下商品圖片或文字按鈕後，使用者可以按下菜單選擇欲購買之份量。分別有大中小份可供買家選擇，價錢也會隨着份量大小使出調整。按下取消按鈕便可重回商品頁面選擇其他產品，否則選擇好份量之後即可按下確認按鈕後加入購物車中。



按下購物車按鍵後的購物車畫面

進入購物車頁面後，可以看見使用者先前在商品頁面所選購的商品及其價錢。顯示所有已加入至購物車的商品及價錢後，系統會計出購物車內所有商品的總金額。購物車同時擁有顯示會員當前的帳號餘額，並計算購買後的所剩餘額。價錢下方要求會員填寫配運資料，會員除了需填寫收貨人姓名之外，還要填寫收貨人手機號碼及收貨地址，並可以選擇是否捐款支持我們的作業。然後會員按下購買按鍵便可成功送出訂單，然後會跳出購買成功提示。



按下地址按鍵後的地圖畫面

使用者可透過按下應用程式左上角菜單或右上角的三點按鈕中選擇地址的話，便會跳出 Google Map 並搜尋相關地點位置。由於目前假想企業沒有實際地址，所以設定了為學校地址。



按下搜索更多按鍵後的 Google 畫面

使用者可透過按下應用程式左上角菜單或右上角的三點按鈕中選擇搜索更多的話，就會跳至 Google Chrome 搜尋所設定的字眼。由於目前假想企業沒有實際內容可搜尋，所以設定了「TaiwanGooGooGood」為搜尋字眼，搜尋有關「國產咕咕 Good」的資訊。



按下留言按鍵後的 Gmail 畫面

使用者可透過按下應用程式左上角菜單或右上角的三點按鈕中選擇留言的話，就會跳至 Gmail 應用程式讓使用者填寫電郵，並留言給企業接收來自顧客的訊息。由於目前假想企業沒有實際電郵地址，跳至 Gmail 應用程式時撰寫電郵時，會預設收件電郵地址為「C110134148@nkust.edut.tw」、預設標題為「Meating Mail Testing」，以及預設內容為「This is Mail Testing.」。



按下關於按鈕後的頁面

使用者可透過按下應用程式左上角菜單或右上角的三點按鈕中選擇關於的話，就會跳至關於頁面顯示企業介紹及相關資料供使用者觀看。由於目前假想企業並沒有相關資訊，目前顯示了組別、專案名稱及組員名單。



電話插入電源顯示提示

c. 功能與相對應的程式片段

```
// Load products data
meat = new HashMap<Integer, Meat>();

try {
    XmlResourceParser xrp = getResources().getXml(R.xml.products);
    xrp.next();

    int eventType = xrp.getEventType();
    String tag = "";
    Meat m = null;

    while (eventType != XmlPullParser.END_DOCUMENT) {
        switch (eventType) {
            case XmlPullParser.START_TAG:
                tag = xrp.getName();

                if (tag.equalsIgnoreCase( anotherString: "product")) {
                    m = new Meat();
                    m.id = xrp.getAttributeValue( namespace: null,  name: "id");
                }
                break;

            case XmlPullParser.TEXT:
                if(!xrp.getText().equals("")) {
                    switch (tag) {
                        case "name":
                            m.name = xrp.getText();
                            break;

                        case "image":
                            m.image = xrp.getText();
                            break;

                        case "priceShort":
                            m.priceShort = Float.parseFloat(xrp.getText());
                            break;

                        case "priceSmall":
                            m.priceSmall = Float.parseFloat(xrp.getText());
                            break;

                        case "priceMedium":
                            m.priceMedium = Float.parseFloat(xrp.getText());
                            break;

                        case "priceLarge":
                            m.priceLarge = Float.parseFloat(xrp.getText());
                            break;

                        default:
                            break;
                    }
                }
                break;

            case XmlPullParser.END_TAG:
                if (xrp.getName().equalsIgnoreCase( anotherString: "product")) {
                    switch (m.id) {
                        case "m1":
                            meat.put(R.id.m1, m);
                            break;
                        case "m2":
                            meat.put(R.id.m2, m);
                            break;
                        case "m3":
                            meat.put(R.id.m3, m);
                            break;
                        case "m4":
                            meat.put(R.id.m4, m);
                            break;
                        case "m5":
                            meat.put(R.id.m5, m);
                            break;
                        case "m6":
                            meat.put(R.id.m6, m);
                            break;
                        case "m7":
                            meat.put(R.id.m7, m);
                            break;
                        default:
                            break;
```



商品頁面相關程式片段

```java
public class CartFragment extends Fragment {

    3 usages
    private FragmentCartBinding binding;

    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {
        CartViewModel cartViewModel =
                new ViewModelProvider( owner: this).get(CartViewModel.class);

        binding = FragmentCartBinding.inflate(inflater, container, attachToParent: false);
        View root = binding.getRoot();

        // 初始化 SharedPreferences
        SharedPreferences sharedPreferences = PreferenceManager.getDefaultSharedPreferences(getContext());

        MainActivity mainActivity = (MainActivity) getActivity();
        String str = "";
        float total = 0.0f;

        for (Map.Entry<Meat, Float> set : mainActivity.selectedMeats.entrySet()) {
            str += set.getKey().name + " $ " + set.getValue() + "\n";
            total += set.getValue();
        }

        float cashAfter = mainActivity.currentCash - total;

        str += "總共: $ " + total + "\n";
        str += "目前餘額: $ " + mainActivity.currentCash + "\n";
        str += "購買後餘額: $ " + cashAfter + "\n";

        ((TextView) root.findViewById(R.id.details)).setText(str);

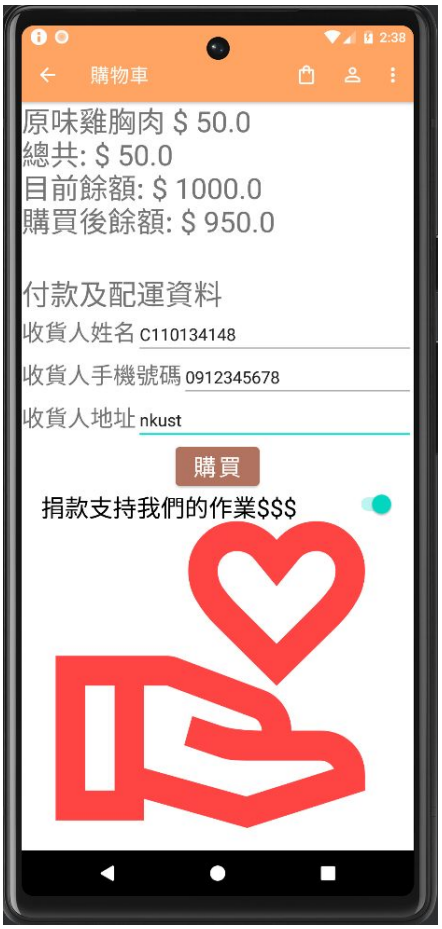
        Switch switchSupport = root.findViewById(R.id.switchSupport);

        boolean switchState = sharedPreferences.getBoolean( key: "switchState",  defValue: false);
        switchSupport.setChecked(switchState);
        String backgroundResource = sharedPreferences.getString( key: "backgroundResource",  defValue: null);
        if (backgroundResource != null) {
            int resourceId = getResources().getIdentifier(backgroundResource,  defType: "drawable", getActivity().getPackageName());
            switchSupport.setBackgroundResource(resourceId);
        } else {
            switchSupport.setBackgroundResource(R.drawable.bg_off);
        }


        switchSupport.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(CompoundButton buttonView, boolean isChecked) {

                if (isChecked) {
                    switchSupport.setBackgroundResource(R.drawable.volunteer_activism_24px);
                    Toast.makeText(getContext(),  text: "感謝您的支持！", Toast.LENGTH_SHORT).show();
                } else {
                    switchSupport.setBackgroundResource(R.drawable.bg_off);
                    Toast.makeText(getContext(),  text: "取消支持", Toast.LENGTH_SHORT).show();
                }
                SharedPreferences.Editor editor = sharedPreferences.edit();
                editor.putBoolean("switchState", isChecked);
                editor.putString("backgroundResource", isChecked ? "volunteer_activism_24px" : "bg_off");
                editor.apply();
            }
        });
        return root;
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
```



購物車相關程式片段 (含 Shared Preferences 功能)



```
public class RecordFragment extends Fragment {

    3 usages
    private FragmentRecordBinding binding;
    2 usages
    private DBHelper dbHelper;

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        RecordViewModel viewModel =
                new ViewModelProvider( owner: this).get(RecordViewModel.class);

        binding = FragmentRecordBinding.inflate(inflater, container,  attachToParent: false);
        View root = binding.getRoot();

        dbHelper = new DBHelper(getContext());

        // 顯示購買記錄
        displayPurchaseRecords(root);

        return root;
    }

    1 usage
    private void displayPurchaseRecords(View root) {
        SQLiteDatabase db = dbHelper.getReadableDatabase();

        String[] projection = {
                DBHelper.COLUMN_ID,
                DBHelper.COLUMN_TOTAL,
                DBHelper.COLUMN_NAME,
                DBHelper.COLUMN_PHONE,
                DBHelper.COLUMN_ADDRESS
        };

        Cursor cursor = db.query(
                DBHelper.TABLE_ORDERS,
                projection,
                 selection: null,
                 selectionArgs: null,
                 groupBy: null,
                having: null,
                DBHelper.COLUMN_ID + " DESC"  // 使用 ORDER BY 子句按 ID 降序排序
        );

        StringBuilder recordsText = new StringBuilder();

        while (cursor.moveToNext()) {
            long orderId = cursor.getLong(cursor.getColumnIndexOrThrow(DBHelper.COLUMN_ID));
            float total = cursor.getFloat(cursor.getColumnIndexOrThrow(DBHelper.COLUMN_TOTAL));
            String name = cursor.getString(cursor.getColumnIndexOrThrow(DBHelper.COLUMN_NAME));
            String phone = cursor.getString(cursor.getColumnIndexOrThrow(DBHelper.COLUMN_PHONE));
            String address = cursor.getString(cursor.getColumnIndexOrThrow(DBHelper.COLUMN_ADDRESS));

            // 在視圖中顯示購買記錄
            String recordText = "訂單編號: " + orderId + "\n" +
                    "總額: $ " + total + "\n" +
                    "收件人姓名: " + name + "\n" +
                    "手機號碼: " + phone + "\n" +
                    "地址: " + address + "\n\n";

            recordsText.append(recordText);
        }

        cursor.close();
        db.close();

        // 將購買記錄添加到 purchase_records 的 TextView 中
        TextView purchaseRecordsTextView = root.findViewById(R.id.purchase_records);
        purchaseRecordsTextView.setText(recordsText.toString());
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
```



```
public class DBHelper extends SQLiteOpenHelper {

    1 usage
    private static final String DATABASE_NAME = "MeatingOrders.db";
    1 usage
    private static final int DATABASE_VERSION = 2;

    // 表名和列名
    4 usages
    public static final String TABLE_ORDERS = "orders";
    2 usages
    public static final String TABLE_USERS = "users"; // 新增的用戶表
    5 usages
    public static final String COLUMN_ID = "_id";
    4 usages
    public static final String COLUMN_TOTAL = "total";
    4 usages
    public static final String COLUMN_NAME = "name";
    4 usages
    public static final String COLUMN_PHONE = "phone";
    4 usages
    public static final String COLUMN_ADDRESS = "address";
    1 usage
    public static final String COLUMN_USER_ID = "user_id"; // 新增的列

    // 創建表的SQL語句
    1 usage
    private static final String TABLE_CREATE_ORDERS =
            "CREATE TABLE " + TABLE_ORDERS + " (" +
                    COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, " +
                    COLUMN_TOTAL + " REAL, " +
                    COLUMN_NAME + " TEXT, " +
                    COLUMN_PHONE + " TEXT, " +
                    COLUMN_ADDRESS + " TEXT, " +
                    COLUMN_USER_ID + " INTEGER);"; // 新增的列

    1 usage
    private static final String TABLE_CREATE_USERS =
            "CREATE TABLE " + TABLE_USERS + " (" +
                    COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, " +
                    "username TEXT UNIQUE NOT NULL, " +
                    "password TEXT NOT NULL);"; // 新增的用戶表

    2 usages
    public DBHelper(Context context) { super(context, DATABASE_NAME,  factory: null, DATABASE_VERSION); }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(TABLE_CREATE_USERS); // 先創建用戶表
        db.execSQL(TABLE_CREATE_ORDERS); // 再創建訂單表
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_ORDERS);
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);

        // 再次創建新的用戶表和訂單表
        onCreate(db);
    }
}
```

購買紀錄相關程式片段＋SQLite Database

```
private void callMap() {
    Uri location = Uri.parse( uriString: "geo:0,0?q=國立高雄科技大學第一校區管理學院");
    Intent mapIntent = new Intent(Intent.ACTION_VIEW, location);
    startActivity(mapIntent);
}
```

地址程式相關片段

```
private void callSearch() {
    Uri webpage = Uri.parse( uriString: "https://www.google.com/search?q=TaiwanGooGooGood");
    Intent webIntent = new Intent(Intent.ACTION_VIEW, webpage);
    startActivity(webIntent);
}
```

搜索更多 程式相關片段

```
private void callMail() {
    Intent emailIntent = new Intent(Intent.ACTION_SEND);

    emailIntent.setType("text/plain");
    emailIntent.putExtra(Intent.EXTRA_EMAIL, new String[] {"C110134148@nkust.edu.tw"}); // recipients
    emailIntent.putExtra(Intent.EXTRA_SUBJECT,  value: "Meating Mail Testing");
    emailIntent.putExtra(Intent.EXTRA_TEXT,  value: "This is Mail Testing.");

    startActivity(emailIntent);
}
```

留言 電郵相關程式片段

```java
public class LoginDiaLogFrament extends DialogFragment {
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());
        // Get the layout inflater
        LayoutInflater inflater = requireActivity().getLayoutInflater();

        // Inflate and set the layout for the dialog
        // Pass null as the parent view because its going in the dialog layout
        View binding = inflater.inflate(R.layout.dialog_signin,  root: null);

        AlertDialog alertDialog = builder.setView(binding)
                .setTitle("Meating")
                // Add action buttons
                .setPositiveButton("登入",  listener: null)
                .setNegativeButton("取消", new DialogInterface.OnClickListener() {
                    public void onClick(DialogInterface dialog, int id) {
                        LoginDiaLogFrament.this.getDialog().cancel();
                    }
                })
                .setNeutralButton("註冊",  listener: null)
                .create();

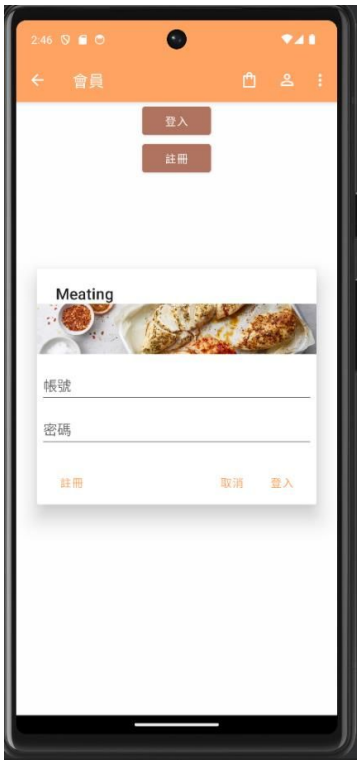
        alertDialog.setOnShowListener(new DialogInterface.OnShowListener() {
            @Override
            public void onShow(DialogInterface dialog) {
                // 登錄按鍵
                Button pBtn = alertDialog.getButton(AlertDialog.BUTTON_POSITIVE);
                pBtn.setOnClickListener(new View.OnClickListener() {
                    @Override
                    public void onClick(View v) {
                        String username = ((TextView) binding.findViewById(R.id.username)).getText().toString();
                        String password = ((TextView) binding.findViewById(R.id.password)).getText().toString();

                        if (checkCredentials(username, password)) {
                            MainActivity mainActivity = (MainActivity) getActivity();
                            mainActivity.onLogInSucceeded();
                            alertDialog.dismiss();
                        } else {
                            Toast.makeText(getContext(),  text: "錯誤帳號或密碼", Toast.LENGTH_SHORT).show();
                        }
                    }
                });
                Button nBtn = alertDialog.getButton(AlertDialog.BUTTON_NEUTRAL);
                nBtn.setOnClickListener(new View.OnClickListener() {
                    @Override
                    public void onClick(View v) {
                        Intent intent = new Intent(getActivity(), RegisterActivity.class);
                        startActivity(intent);
                        alertDialog.dismiss();
                    }
                });

            }
        });

        return alertDialog;
    }
    1 usage
    private boolean checkCredentials(String username, String password) {
        SharedPreferences preferences = getActivity().getSharedPreferences( name: "user_data", Context.MODE_PRIVATE);
        String storedUsername = preferences.getString( key: "username",  defValue: "");
        String storedPassword = preferences.getString( key: "password",  defValue: "");

        return username.equals(storedUsername) && password.equals(storedPassword);
    }
}
```



會員登入程式相關片段(含 Shared Preferences 功能)

```java
public class AddValueFragment extends Fragment {
    4 usages
    public static AddValueFragment instance;
    4 usages
    private FragmentAddvalueBinding binding;

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        instance = this;

        AddValueViewModel viewModel =
                new ViewModelProvider( owner: this).get(AddValueViewModel.class);

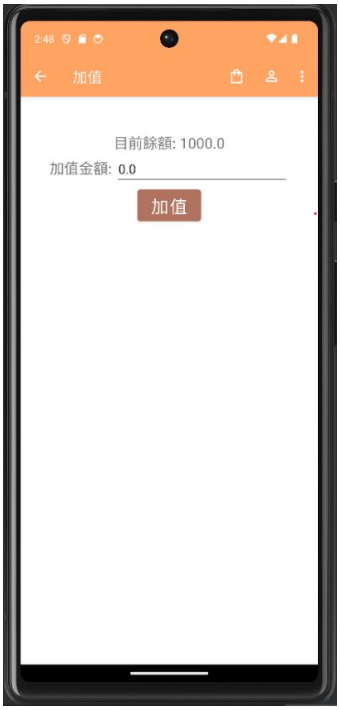
        binding = FragmentAddvalueBinding.inflate(inflater, container, attachToParent: false);
        View root = binding.getRoot();

        MainActivity mainActivity = (MainActivity) getActivity();
        ((TextView) root.findViewById(R.id.currentCash)).setText("目前餘額: " + mainActivity.currentCash);
        ((EditText) root.findViewById(R.id.input_amount)).setText("0.0");

        return root;
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
        instance = null;
    }

    2 usages
    public float getAddAmount() {
        String str = ((EditText) binding.getRoot().findViewById(R.id.input_amount)).getText().toString();
        return Float.parseFloat(str);
    }
}
```



加值功能相關程式片段

```java
public class CustomReceiver extends BroadcastReceiver {
    no usages
    private static final String ACTION_CUSTOM_BROADCAST =
            BuildConfig.APPLICATION_ID + ".ACTION_CUSTOM_BROADCAST";

    @Override
    public void onReceive(Context context, Intent intent) {
        String intentAction = intent.getAction();

        if (intentAction != null) {
            String toastMessage = "unknown intent action";
            switch (intentAction) {
                case Intent.ACTION_POWER_CONNECTED:
                    toastMessage = "充電了！";
                    break;
                case Intent.ACTION_POWER_DISCONNECTED:
                    toastMessage = "電源已斷開！";
                    break;
            }

            // Display the toast.
            Toast.makeText(context, toastMessage, Toast.LENGTH_SHORT).show();
        }
    }
}
```





BroadcastReceiver 功能

二.參考文獻

圖片來源：

https://www.facebook.com/TaiwanGooGooGood/

https://www.bbc.co.uk/food/recipes/baked_chicken_breast_49731

https://tw.buy.yahoo.com/gdsale/gdsale.asp?gdid=p0797236083295&from=pla&gg=0&co_servername=rsPmax2.0_dashop188

程式參考文獻：

https://evanchen76.medium.com/android-ui-%E4%BD%88%E5%B1%80-part6-scrollview-c84d614288f3

https://juejin.cn/post/7103525485828636686

三.附錄(列出重要功能)

MainActivity.java

```
public class MainActivity extends AppCompatActivity implements LogInOutListener {

    private AppBarConfiguration mAppBarConfiguration;
    private ActivityMainBinding binding;

    public boolean hasBeenLoggedIn;
    public HashMap<Integer, Meat> meat;
    public Meat meatForProductDialog;
    public LinkedHashMap<Meat, Float> selectedMeats;
    public ArrayList<AbstractMap.SimpleEntry<Long, Float>> records;
    public float currentCash = 1000.0f;

    private CustomReceiver mReceiver = new CustomReceiver();
    private static final String ACTION_CUSTOM_BROADCAST =
            BuildConfig.APPLICATION_ID + ".ACTION_CUSTOM_BROADCAST";


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        //setContentView(R.layout.activity_main);

        hasBeenLoggedIn = false;
        selectedMeats = new LinkedHashMap<Meat, Float>();
        records = new ArrayList<AbstractMap.SimpleEntry<Long, Float>>();

        binding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());

        setSupportActionBar(binding.appBarMain.toolbar);

        DrawerLayout drawer = binding.drawerLayout;
        NavigationView navigationView = binding.navView;

        IntentFilter filter = new IntentFilter();
        filter.addAction(Intent.ACTION_POWER_DISCONNECTED);
        filter.addAction(Intent.ACTION_POWER_CONNECTED);

        // Register the receiver using the activity context.
        this.registerReceiver(mReceiver, filter);

        LocalBroadcastManager.getInstance(this)
                .registerReceiver(mReceiver,
                        new IntentFilter(ACTION_CUSTOM_BROADCAST));

        // Passing each menu ID as a set of Ids because each
        // menu should be considered as top level destinations.
        mAppBarConfiguration = new AppBarConfiguration.Builder(
                R.id.nav_home)
                .setOpenableLayout(drawer)
                .build();
        NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
        NavigationUI.setupActionBarWithNavController(this, navController, mAppBarConfiguration);
        NavigationUI.setupWithNavController(navigationView, navController);

        // Must be put after setupWithNavController
        navigationView.setNavigationItemSelectedListener(new NavigationView.OnNavigationItemSelectedListener() {
            @Override
            public boolean onNavigationItemSelected(MenuItem item) {
                int id = 0;
```

```
        if (item.getItemId() == R.id.nav_map) id = 1;
        if (item.getItemId() == R.id.action_search) id = 2;
        if (item.getItemId() == R.id.action_comment) id = 3;
        switch (id) {
            case 1:
                callMap();
                break;
            case 2:
                callSearch();
                break;
            case 3:
                callMail();
                break;
            default:
                NavigationUI.onNavDestinationSelected(item, navController);
                break;
        }

        drawer.close();

        return true;
    }
});

// Load products data
meat = new HashMap<Integer, Meat>();

try {
    XmlResourceParser xrp = getResources().getXml(R.xml.products);
    xrp.next();

    int eventType = xrp.getEventType();
    String tag = "";
    Meat m = null;

    while (eventType != XmlPullParser.END_DOCUMENT) {
        switch (eventType) {
            case XmlPullParser.START_TAG:
                tag = xrp.getName();

                if (tag.equalsIgnoreCase("product")) {
                    m = new Meat();
                    m.id = xrp.getAttributeValue(null, "id");
                }
                break;

            case XmlPullParser.TEXT:
                if (!xrp.getText().equals("")) {
                    switch (tag) {
                        case "name":
                            m.name = xrp.getText();
                            break;

                        case "image":
                            m.image = xrp.getText();
                            break;

                        case "priceShort":
                            m.priceShort = Float.parseFloat(xrp.getText());
                            break;

                        case "priceSmall":
                            m.priceSmall = Float.parseFloat(xrp.getText());
                            break;

                        case "priceMedium":
```

```
                        m.priceMedium = Float.parseFloat(xrp.getText());
                        break;

                    case "priceLarge":
                        m.priceLarge = Float.parseFloat(xrp.getText());
                        break;

                    default:
                        break;
                }
            }
            break;

        case XmlPullParser.END_TAG:
            if (xrp.getName().equalsIgnoreCase("product")) {
                switch (m.id) {
                    case "m1":
                        meat.put(R.id.m1, m);
                        break;
                    case "m2":
                        meat.put(R.id.m2, m);
                        break;
                    case "m3":
                        meat.put(R.id.m3, m);
                        break;
                    case "m4":
                        meat.put(R.id.m4, m);
                        break;
                    case "m5":
                        meat.put(R.id.m5, m);
                        break;
                    case "m6":
                        meat.put(R.id.m6, m);
                        break;
                    case "m7":
                        meat.put(R.id.m7, m);
                        break;
                    default:
                        break;
                }
            }
            break;

        default:
            break;
      }

      eventType = xrp.next();
    }
  } catch (IOException e) {
    e.printStackTrace();
  } catch (XmlPullParserException e) {
    e.printStackTrace();
  }
}

@Override
public void onStart() {
  super.onStart();
  Toast.makeText(this, "Meating 歡迎您!", Toast.LENGTH_SHORT).show();
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
  // Inflate the menu; this adds items to the action bar if it is present.
  getMenuInflater().inflate(R.menu.main, menu);
```

```
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = 0;
    if (item.getItemId() == R.id.nav_map) id = 1;
    if (item.getItemId() == R.id.action_search) id = 2;
    if (item.getItemId() == R.id.action_comment) id = 3;
    switch (item.getItemId()) {
        case 1:
            callMap();
            return true;
        case 2:
            callSearch();
            return true;
        case 3:
            callMail();
            break;
        default:
            NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
            return NavigationUI.onNavDestinationSelected(item, navController)
                    || super.onOptionsItemSelected(item);
    }

    return true;
}

@Override
public boolean onSupportNavigateUp() {
    NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
    return NavigationUI.navigateUp(navController, mAppBarConfiguration)
            || super.onSupportNavigateUp();
}

public void logIn() {
    DialogFragment newFragment = new LoginDiaLogFrament();
    newFragment.show(getSupportFragmentManager(), "Login");
}

public void logOut() {
    onLogOutSucceeded();
}

@Override
public void onLogInSucceeded() {
    hasBeenLoggedIn = true;

    if (MemberFragment.instance != null)
        MemberFragment.instance.onLogOutSucceeded();
    // 登录成功后显示通知
    showLoginNotification("登入成功", "歡迎您回來！");

}

@Override
public void onLogOutSucceeded() {
    hasBeenLoggedIn = false;

    if (MemberFragment.instance != null)
        MemberFragment.instance.onLogOutSucceeded();

    // 導航回主頁
    NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
    navController.navigate(R.id.nav_home);
}
```

```
public void onBtnClick(View view) {
    int id = 0;
    if (view.getId() == R.id.btn_logIn) id = 1;
    if (view.getId() == R.id.btn_signUp) id = 2;
    if (view.getId() == R.id.btn_logOut) id = 3;
    switch (id) {
        case 1:
            logIn();
            break;

        case 2:
            Intent intent = new Intent(this, RegisterActivity.class);
            startActivity(intent);
            break;

        case 3:
            logOut();
            break;

        default:
            break;
    }
}

public void onClickAddValue(View view) {
    NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
    navController.navigate(R.id.nav_addvalue);
}

public void onClickRecord(View view) {
    NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
    navController.navigate(R.id.nav_record);
}

public void onClickPersonal(View view) {
    NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
    navController.navigate(R.id.nav_personal);
}

public void onClickProduct(View view) {
    meatForProductDialog = meat.get(view.getId());
    ProductDialogFragment newFragment = new ProductDialogFragment();
    newFragment.show(getSupportFragmentManager(), "Product");
}

public void callShop() {
    NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
    navController.navigate(R.id.nav_shop);
}

private void callMap() {
    Uri location = Uri.parse("geo:0,0?q=國立高雄科技大學第一校區管理學院");
    Intent mapIntent = new Intent(Intent.ACTION_VIEW, location);
    startActivity(mapIntent);
}

private void callSearch() {
    Uri webpage = Uri.parse("https://www.google.com/search?q=TaiwanGooGooGood");
    Intent webIntent = new Intent(Intent.ACTION_VIEW, webpage);
    startActivity(webIntent);
}

private void callMail() {
    Intent emailIntent = new Intent(Intent.ACTION_SEND);
```

```
        emailIntent.setType("text/plain");
        emailIntent.putExtra(Intent.EXTRA_EMAIL, new String[]{"C110134148@nkust.edu.tw"}); // recipients
        emailIntent.putExtra(Intent.EXTRA_SUBJECT, "Meating Mail Testing");
        emailIntent.putExtra(Intent.EXTRA_TEXT, "This is Mail Testing.");

        startActivity(emailIntent);
    }

    public void onPurchaseBtnClick(View view) {
        float total = 0.0f;

        for (Map.Entry<Meat, Float> set : selectedMeats.entrySet()) {
            total += set.getValue();
        }

        float cashAfter = currentCash - total;

        if (cashAfter < 0.0f) {
            Toast.makeText(this, "購買失敗", Toast.LENGTH_SHORT).show();
        } else {
            currentCash -= total;

            // 插入購買記錄到 SQLite 數據庫
            DBHelper dbHelper = new DBHelper(this);
            SQLiteDatabase db = dbHelper.getWritableDatabase();
            EditText editTextName = findViewById(R.id.editTextTextPersonName);
            EditText editTextPhone = findViewById(R.id.editTextPhone);
            EditText editTextAddress = findViewById(R.id.editTextTextPostalAddress);
            ContentValues values = new ContentValues();
            values.put(DBHelper.COLUMN_TOTAL, total);
            values.put(DBHelper.COLUMN_NAME, editTextName.getText().toString());
            values.put(DBHelper.COLUMN_PHONE, editTextPhone.getText().toString());
            values.put(DBHelper.COLUMN_ADDRESS, editTextAddress.getText().toString());

            long newRowId = db.insert(DBHelper.TABLE_ORDERS, null, values);

            db.close();

            Date d = new Date();
            records.add(new AbstractMap.SimpleEntry<>(d.getTime(), total));
            selectedMeats.clear();
            NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
            navController.navigate(R.id.nav_home);

            // 显示购买成功的通知
            showPurchaseNotification("購買成功", "您已成功購買產品");
        }
    }


    private void showPurchaseNotification(String title, String message) {
        NotificationManager notificationManager = (NotificationManager) getSystemService(Context.NOTIFICATION_SERVICE);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            NotificationChannel channel = new NotificationChannel("purchase_channel", "Purchase Channel",
NotificationManager.IMPORTANCE_HIGH);
            notificationManager.createNotificationChannel(channel);
        }

        NotificationCompat.Builder builder = new NotificationCompat.Builder(this, "purchase_channel")
                .setSmallIcon(android.R.drawable.ic_dialog_info)
                .setContentTitle(title)
                .setContentText(message)
                .setPriority(NotificationCompat.PRIORITY_HIGH);
```

```
        notificationManager.notify(1, builder.build());
    }

    private void showLoginNotification(String title, String message) {
        NotificationManager notificationManager = (NotificationManager) getSystemService(Context.NOTIFICATION_SERVICE);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            NotificationChannel channel = new NotificationChannel("login_channel", "Login Channel", NotificationManager.IMPORTANCE_HIGH);
            notificationManager.createNotificationChannel(channel);
        }

        NotificationCompat.Builder builder = new NotificationCompat.Builder(this, "login_channel")
                .setSmallIcon(android.R.drawable.ic_dialog_info)
                .setContentTitle(title)
                .setContentText(message)
                .setPriority(NotificationCompat.PRIORITY_HIGH);

        notificationManager.notify(3, builder.build());
    }

    public void onAddvalueClick(View view) {
        if (AddValueFragment.instance.getAddAmount() > 0.0f) {
            currentCash += AddValueFragment.instance.getAddAmount();
            NavController navController = Navigation.findNavController(this, R.id.nav_host_fragment_content_main);
            navController.navigate(R.id.nav_member);
            Toast.makeText(this, "加值成功", Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "錯誤數值", Toast.LENGTH_SHORT).show();
        }
    }

    public void sendCustomBroadcast(View view) {
        Intent customBroadcastIntent = new Intent(ACTION_CUSTOM_BROADCAST);
        LocalBroadcastManager.getInstance(this).sendBroadcast(customBroadcastIntent);

    }


    @Override
    protected void onDestroy() {
        // Unregister the receiver.
        this.unregisterReceiver(mReceiver);
        super.onDestroy();
        LocalBroadcastManager.getInstance(this)
                .unregisterReceiver(mReceiver);
    }
}
```

ProductDialogFragment.java

```
package com.example.meating;

import android.app.Dialog;
import android.content.DialogInterface;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.DialogFragment;
import com.example.meating.ui.shop.Meat;
```

```
public class ProductDialogFragment extends DialogFragment {
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        MainActivity mainActivity = (MainActivity) getActivity();
        Meat m = mainActivity.meatForProductDialog;
        AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());
        // Get the layout inflater
        LayoutInflater inflater = requireActivity().getLayoutInflater();
        // Inflate and set the layout for the dialog
        // Pass null as the parent view because its going in the dialog layout
        View binding = inflater.inflate(R.layout.dialog_product, null);
        Spinner spinner = (Spinner) binding.findViewById(R.id.p_size);
        ArrayAdapter<CharSequence> adapter = ArrayAdapter.createFromResource(getContext(),
                R.array.sizes_array, android.R.layout.simple_spinner_item);
        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
        spinner.setAdapter(adapter);
        spinner.setSelected(true);
        ((TextView)binding.findViewById(R.id.p_price)).setText("$" + Float.toString(m.priceSmall));
        spinner.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
            @Override
            public void onItemSelected(AdapterView<?> parent, View view, int position, long id) {
                switch(position) {
                    case 0:
                        ((TextView)binding.findViewById(R.id.p_price)).setText("$" + Float.toString(m.priceSmall));
                        break;
                    case 1:
                        ((TextView)binding.findViewById(R.id.p_price)).setText("$" + Float.toString(m.priceMedium));
                        break;
                    case 2:
                        ((TextView)binding.findViewById(R.id.p_price)).setText("$" + Float.toString(m.priceLarge));
                        break;
                }
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent) {
            }
        });
        AlertDialog alertDialog = builder.setView(binding)
                .setTitle(m.name)
                // Add action buttons
                .setPositiveButton(R.string.OK, new DialogInterface.OnClickListener() {
                    public void onClick(DialogInterface dialog, int id) {
                        MainActivity mainActivity = (MainActivity) getActivity();

                        switch(spinner.getSelectedItemPosition()) {
                            case 0:
                                mainActivity.selectedMeats.put(m, m.priceSmall);
                                break;
                            case 1:
                                mainActivity.selectedMeats.put(m, m.priceMedium);
                                break;
                            case 2:
                                mainActivity.selectedMeats.put(m, m.priceLarge);
                                break;
                        }
                    }
                })
                .setNegativeButton(R.string.cancel, new DialogInterface.OnClickListener() {
                    public void onClick(DialogInterface dialog, int id) {
                        ProductDialogFragment.this.getDialog().cancel();
                    }
                })
                .create();
        return alertDialog;
    }
}
```

DBHelper.java

```
package com.example.meating;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "MeatingOrders.db";
    private static final int DATABASE_VERSION = 2;

    // 表名和列名
    public static final String TABLE_ORDERS = "orders";
    public static final String TABLE_USERS = "users"; // 新增的用戶表
    public static final String COLUMN_ID = "_id";
    public static final String COLUMN_TOTAL = "total";
    public static final String COLUMN_NAME = "name";
    public static final String COLUMN_PHONE = "phone";
    public static final String COLUMN_ADDRESS = "address";
    public static final String COLUMN_USER_ID = "user_id"; // 新增的列

    // 創建表的 SQL 語句
    private static final String TABLE_CREATE_ORDERS =
            "CREATE TABLE " + TABLE_ORDERS + " (" +
                    COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT," +
                    COLUMN_TOTAL + " REAL," +
                    COLUMN_NAME + " TEXT," +
                    COLUMN_PHONE + " TEXT," +
                    COLUMN_ADDRESS + " TEXT," +
                    COLUMN_USER_ID + " INTEGER)"; // 新增的列

    private static final String TABLE_CREATE_USERS =
            "CREATE TABLE " + TABLE_USERS + " (" +
                    COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT," +
                    "username TEXT UNIQUE NOT NULL," +
                    "password TEXT NOT NULL);"; // 新增的用戶表

    public DBHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(TABLE_CREATE_USERS); // 先創建用戶表
        db.execSQL(TABLE_CREATE_ORDERS); // 再創建訂單表
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_ORDERS);
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);

        // 再次創建新的用戶表和訂單表
        onCreate(db);
    }
}
```

CustomReceiver.java

```
package com.example.meating;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;
//import com.example.meating.BuildConfig;


public class CustomReceiver extends BroadcastReceiver {
    private static final String ACTION_CUSTOM_BROADCAST =
            BuildConfig.APPLICATION_ID + ".ACTION_CUSTOM_BROADCAST";

    @Override
    public void onReceive(Context context, Intent intent) {
        String intentAction = intent.getAction();

        if (intentAction != null) {
            String toastMessage = "unknown intent action";
            switch (intentAction) {
                case Intent.ACTION_POWER_CONNECTED:
                    toastMessage = "充電了！";
                    break;
                case Intent.ACTION_POWER_DISCONNECTED:
                    toastMessage = "電源已斷開！";
                    break;
            }

            // Display the toast.
            Toast.makeText(context, toastMessage, Toast.LENGTH_SHORT).show();
        }
    }
}
```

LoginDialogFragment.java

```
package com.example.meating;

import android.app.Dialog;
import android.content.Context;
import android.content.DialogInterface;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.DialogFragment;

public class LoginDiaLogFrament extends DialogFragment {
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());
        // Get the layout inflater
        LayoutInflater inflater = requireActivity().getLayoutInflater();
```

```
        // Inflate and set the layout for the dialog
        // Pass null as the parent view because its going in the dialog layout
        View binding = inflater.inflate(R.layout.dialog_signin, null);

        AlertDialog alertDialog = builder.setView(binding)
                .setTitle("Meating")
                // Add action buttons
                .setPositiveButton(R.string.login, null)
                .setNegativeButton(R.string.cancel, new DialogInterface.OnClickListener() {
                    public void onClick(DialogInterface dialog, int id) {
                        LoginDiaLogFrament.this.getDialog().cancel();
                    }
                })
                .setNeutralButton(R.string.register, null)
                .create();

        alertDialog.setOnShowListener(new DialogInterface.OnShowListener() {
            @Override
            public void onShow(DialogInterface dialog) {
                // 登錄按鈕
                Button pBtn = alertDialog.getButton(AlertDialog.BUTTON_POSITIVE);
                pBtn.setOnClickListener(new View.OnClickListener() {
                    @Override
                    public void onClick(View v) {
                        String username = ((TextView) binding.findViewById(R.id.username)).getText().toString();
                        String password = ((TextView) binding.findViewById(R.id.password)).getText().toString();

                        if (checkCredentials(username, password)) {
                            MainActivity mainActivity = (MainActivity) getActivity();
                            mainActivity.onLogInSucceeded();
                            alertDialog.dismiss();
                        } else {
                            Toast.makeText(getContext(), "錯誤帳號或密碼", Toast.LENGTH_SHORT).show();
                        }
                    }
                });
                Button nBtn = alertDialog.getButton(AlertDialog.BUTTON_NEUTRAL);
                nBtn.setOnClickListener(new View.OnClickListener() {
                    @Override
                    public void onClick(View v) {
                        Intent intent = new Intent(getActivity(), RegisterActivity.class);
                        startActivity(intent);
                        alertDialog.dismiss();
                    }
                });

            }
        });

        return alertDialog;
    }
    private boolean checkCredentials(String username, String password) {
        SharedPreferences preferences = getActivity().getSharedPreferences("user_data", Context.MODE_PRIVATE);
        String storedUsername = preferences.getString("username", "");
        String storedPassword = preferences.getString("password", "");

        return username.equals(storedUsername) && password.equals(storedPassword);
    }
}
```

MemberFragment.java

```
package com.example.meating.ui.member;

import android.os.Bundle;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.ViewModelProvider;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Toast;

import com.example.meating.MainActivity;
import com.example.meating.LogInOutListener;
import com.example.meating.databinding.FragmentMemberBinding;
public class MemberFragment extends Fragment implements LogInOutListener  {

    static public MemberFragment instance;
    private FragmentMemberBinding binding;

    @Override
    public void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        instance = this;
    }

    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             ViewGroup container, Bundle savedInstanceState) {
        MemberViewModel viewModel =
                new ViewModelProvider(this).get(MemberViewModel.class);

        binding = FragmentMemberBinding.inflate(inflater, container, false);
        View root = binding.getRoot();

        return root;
    }

    @Override
    public void onStart() {
        super.onStart();
        setBtnsStatus();
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
        instance = null;
    }

    @Override
    public void onLogInSucceeded() {
        setBtnsStatus();
    }

    @Override
    public void onLogOutSucceeded() {
        setBtnsStatus();
    }
```

```
    public void setBtnsStatus() {
        MainActivity mainActivity = (MainActivity) getActivity();

        if(!mainActivity.hasBeenLoggedIn) {
            binding.btnLogIn.setVisibility(View.VISIBLE);
            binding.btnSignUp.setVisibility(View.VISIBLE);
            binding.navAddvalue.setVisibility(View.GONE);
            binding.navRecord.setVisibility(View.GONE);
            binding.navPersonal.setVisibility(View.GONE);
            binding.btnLogOut.setVisibility(View.GONE);
        } else {
            binding.btnLogIn.setVisibility(View.GONE);
            binding.btnSignUp.setVisibility(View.GONE);
            binding.navAddvalue.setVisibility(View.VISIBLE);
            binding.navRecord.setVisibility(View.VISIBLE);
            binding.navPersonal.setVisibility(View.VISIBLE);
            binding.btnLogOut.setVisibility(View.VISIBLE);
            Toast.makeText(getContext(), "Welcome back!", Toast.LENGTH_SHORT).show();
        }
    }
}
```

CartFragment.java

```
package com.example.meating.ui.cart;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.preference.PreferenceManager;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.CompoundButton;
import android.widget.EditText;
import android.widget.Switch;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.lifecycle.ViewModelProvider;



import com.example.meating.MainActivity;
import com.example.meating.R;
import com.example.meating.databinding.FragmentCartBinding;
import com.example.meating.ui.shop.Meat;

import java.util.Map;

public class CartFragment extends Fragment {

    private FragmentCartBinding binding;

    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {
        CartViewModel cartViewModel =
                new ViewModelProvider(this).get(CartViewModel.class);

        binding = FragmentCartBinding.inflate(inflater, container, false);
```

```
        View root = binding.getRoot();

        // 初始化 SharedPreferences
        SharedPreferences sharedPreferences = PreferenceManager.getDefaultSharedPreferences(getContext());

        MainActivity mainActivity = (MainActivity) getActivity();
        String str = "";
        float total = 0.0f;

        for (Map.Entry<Meat, Float> set : mainActivity.selectedMeats.entrySet()) {
            str += set.getKey().name + " $ " + set.getValue() + "\n";
            total += set.getValue();
        }

        float cashAfter = mainActivity.currentCash - total;

        str += "總共: $ " + total + "\n";
        str += "目前餘額: $ " + mainActivity.currentCash + "\n";
        str += "購買後餘額: $ " + cashAfter + "\n";

        ((TextView) root.findViewById(R.id.details)).setText(str);

        Switch switchSupport = root.findViewById(R.id.switchSupport);

        boolean switchState = sharedPreferences.getBoolean("switchState", false);
        switchSupport.setChecked(switchState);
        String backgroundResource = sharedPreferences.getString("backgroundResource", null);
        if (backgroundResource != null) {
            int resourceId = getResources().getIdentifier(backgroundResource, "drawable", getActivity().getPackageName());
            switchSupport.setBackgroundResource(resourceId);
        } else {
            switchSupport.setBackgroundResource(R.drawable.bg_off);
        }


        switchSupport.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(CompoundButton buttonView, boolean isChecked) {

                if (isChecked) {
                    switchSupport.setBackgroundResource(R.drawable.volunteer_activism_24px);
                    Toast.makeText(getContext(), "感謝您的支持！", Toast.LENGTH_SHORT).show();
                } else {
                    switchSupport.setBackgroundResource(R.drawable.bg_off);
                    Toast.makeText(getContext(), "取消支持", Toast.LENGTH_SHORT).show();
                }
                SharedPreferences.Editor editor = sharedPreferences.edit();
                editor.putBoolean("switchState", isChecked);
                editor.putString("backgroundResource", isChecked ? "volunteer_activism_24px" : "bg_off");
                editor.apply();
            }
        });
        return root;
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}
```

Fragment_shop.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ui.shop.ShopFragment">

    <androidx.core.widget.NestedScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent" >

        <LinearLayout
            android:id="@+id/layout_products"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical">

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="vertical">

                <!-- m1 ImageButton with TextView on the same line -->
                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="horizontal">

                    <ImageButton
                        android:id="@+id/m1"
                        android:layout_width="250dp"
                        android:layout_height="150dp"
                        android:layout_margin="4dp"
                        android:backgroundTint="#FFE2D8"
                        android:onClick="onClickProduct"
                        android:scaleType="center"
                        android:src="@drawable/classicchicken" />

                    <TextView
                        android:layout_width="wrap_content"
                        android:layout_height="wrap_content"
                        android:layout_gravity="center"
                        android:gravity="center"
                        android:text="\t 即食雞胸肉\n 原味"
                        android:textSize="24sp" />
                </LinearLayout>

                <!-- m2 ImageButton with TextView on the next line -->
                <LinearLayout
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:orientation="horizontal">

                    <ImageButton
                        android:id="@+id/m2"
                        android:layout_width="250dp"
                        android:layout_height="150dp"
                        android:layout_margin="4dp"
                        android:backgroundTint="#FFE2D8"
                        android:onClick="onClickProduct"
                        android:scaleType="center"
                        android:src="@drawable/saladchicken" />
```

```
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:text="\t 即食雞胸肉\n 沙拉"
            android:textSize="24sp"
            android:gravity="center"/>
    </LinearLayout>

    <!-- m3 ImageButton with TextView on the next line -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <ImageButton
            android:id="@+id/m3"
            android:layout_width="250dp"
            android:layout_height="150dp"
            android:layout_margin="4dp"
            android:backgroundTint="#FFE2D8"
            android:onClick="onClickProduct"
            android:src="@drawable/meat" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:text="\t 肋眼大牛排\n 黑胡椒"
            android:textSize="24sp"
            android:gravity="center" />
    </LinearLayout>

</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical">

    <!-- ... (m1, m2, m3 的 LinearLayout) ... -->

    <!-- m4 ImageButton with TextView on the same line -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <ImageButton
            android:id="@+id/m4"
            android:layout_width="250dp"
            android:layout_height="150dp"
            android:layout_margin="4dp"
            android:backgroundTint="#FFE2D8"
            android:onClick="onClickProduct"
            android:src="@drawable/friedchicken" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:text="\t 冷凍鹽酥雞\n 辣味"
            android:textSize="24sp"
            android:gravity="center" />
```

```
    </LinearLayout>

    <!-- m5 ImageButton with TextView on the next line -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <ImageButton
            android:id="@+id/m5"
            android:layout_width="250dp"
            android:layout_height="150dp"
            android:layout_margin="4dp"
            android:backgroundTint="#FFE2D8"
            android:onClick="onClickProduct"
            android:src="@drawable/winechicken2" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:text="\t 即食大雞捲\n 紹興酒"
            android:textSize="24sp"
            android:gravity="center"  />
    </LinearLayout>

    <!-- m6 ImageButton with TextView on the next line -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <ImageButton
            android:id="@+id/m6"
            android:layout_width="250dp"
            android:layout_height="150dp"
            android:layout_margin="4dp"
            android:backgroundTint="#FFE2D8"
            android:onClick="onClickProduct"
            android:src="@drawable/mcnuggets" />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            android:text="\t 冷凍麥樂雞\n 原味"
            android:textSize="24sp"
            android:gravity="center"  />
    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <ImageButton
            android:id="@+id/m7"
            android:layout_width="250dp"
            android:layout_height="150dp"
            android:layout_margin="4dp"
            android:backgroundTint="#FFE2D8"
            android:onClick="onClickProduct"
            android:src="@drawable/chickenfinger" />

        <TextView
            android:layout_width="wrap_content"
```

```
                android:layout_height="wrap_content"
                android:layout_gravity="center"
                android:text="\t 冷凍炸雞指\n 胡麻醬"
                android:textSize="24sp"
                android:gravity="center"  />
            </LinearLayout>
        </LinearLayout>
    </LinearLayout>
  </androidx.core.widget.NestedScrollView>
  <com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab_checkout"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_marginRight="36dp"
    android:layout_marginBottom="64dp"
    android:clickable="true"
    android:src="@drawable/shopping_cart_24px"
    app:backgroundTint="#FFA462" />

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

Dialog_signin.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content">
  <ImageView
    android:layout_width="match_parent"
    android:layout_height="64dp"
    android:background="#FFFFBB33"
    android:contentDescription="@string/app_name"
    android:scaleType="centerCrop"
    android:src="@drawable/chickenlogin" />
  <EditText
    android:id="@+id/username"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginLeft="4dp"
    android:layout_marginTop="16dp"
    android:layout_marginRight="4dp"
    android:layout_marginBottom="4dp"
    android:hint="@string/username"
    android:inputType="textEmailAddress"
    android:minHeight="48dp"
    android:autofillHints="" />
  <EditText
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginLeft="4dp"
    android:layout_marginTop="4dp"
    android:layout_marginRight="4dp"
    android:layout_marginBottom="16dp"
    android:fontFamily="sans-serif"
    android:hint="@string/password"
    android:inputType="textPassword"
    android:minHeight="48dp"
    android:autofillHints="" />
</LinearLayout>
```

Fragment_cart.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true"
    tools:context=".MainActivity">

    <LinearLayout
        android:id="@+id/linearLayout"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">

        <TextView
            android:id="@+id/details"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Details"
            android:textSize="30sp" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <TextView
                android:id="@+id/textView"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:gravity="center"
                android:text="付款及配運資料"
                android:textSize="30dp" />

        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <TextView
                android:id="@+id/textView2"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="收貨人姓名"
                android:textSize="24sp" />

            <EditText
                android:id="@+id/editTextTextPersonName"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:ems="10"
                android:inputType="textPersonName" />
        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <TextView
```

```
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="收貨人手機號碼"
        android:textSize="24sp" />

    <EditText
        android:id="@+id/editTextPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="phone" />
  </LinearLayout>

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="收貨人地址"
        android:textSize="24sp" />

    <EditText
        android:id="@+id/editTextTextPostalAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPostalAddress" />
  </LinearLayout>

  <Button
    android:id="@+id/btn_purchase"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center_vertical|center_horizontal"
    android:backgroundTint="#B1735F"
    android:onClick="onPurchaseBtnClick"
    android:text="購買"
    android:textSize="24sp" />

  <Switch
    android:id="@+id/switchSupport"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_marginStart="20dp"
    android:layout_marginEnd="20dp"
    android:text="捐款支持我們的作業$$$"
    android:textSize="25sp"
    android:gravity="top"/>
</LinearLayout>
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```